Bhargav Limbani **92301733029**

Take Home Test-2

Aim :

Complete the following task and submit pdf with output screenshot and zip with all source code:

Sr. No.	Section	Task Description	CO Mapped
1	Backend API (Node + Express)	Set up Node.js + Express project with clean architecture folder structure (src/domain/entities, src/application/use-cases, src/infrastructure, src/presentation/controllers & routes), install dependencies (express, dotenv, jsonwebtoken, bcryptjs, winston, helmet, cors, express-validator), and create server entry with global logging middleware (Winston) and security headers (Helmet).	2, 3
2	Backend API (Node + Express)	Implement authentication module: define User entity in domain, create Register/Login use-cases in application layer, auth controller/routes with JWT generation, bcrypt hashing, role assignment, input validation, and Winston logging for auth events; add JWT middleware for protected routes.	2
3	Backend API (Node + Express)	Develop Book management module: create Book entity, CRUD use-cases (create/read/update/delete/search), controllers and routes with librarian-only role middleware, express-validator, and detailed Winston logging for each operation.	2, 3
4	Backend API (Node + Express)	Implement User management module: use-cases for CRUD/profile operations (restricted to librarian), controllers/routes with RBAC middleware, input sanitization, and logging for user actions.	2
5	Backend API (Node + Express)	Create Transaction management module: define Transaction entity, use-cases for issue (availability check + due date = 14 days), renew (extend due date), return (update status + fine calculation if overdue), controllers/routes with role checks and comprehensive Winston transaction logging.	2, 3
6	Backend API (Node + Express)	Build Reports module: use-cases for various reports (overdue books, popular books, user-wise history, inventory summary) using business logic, protected librarian-only controllers/routes, and logging for report generation.	2

Output :

POST http://localhost:3000/a POST http://localhost:3000/a No environment

http://localhost:3000/api/auth/register Save Share

POST http://localhost:3000/api/auth/register Send

Docs Params Auth Headers (8) Body Scripts Settings Cookies

raw JSON Schema Beautify

```
{
  "name": "Admin Librarian",
  "email": "bhargav@gmail.com",
  "password": "password123",
  "role": "librarian"
}
```

Body 201 Created 899 ms 1.05 KB

{ } JSON Preview Visualize

```
{
    "message": "User registered successfully",
    "user": {
        "id": "48d1972c-8b3e-4d37-9e10-cda856f9231e",
        "name": "Admin Librarian",
        "email": "bhargav@gmail.com",
        "role": "librarian"
    }
}
```

POST http://localhost:3000/a POST http://localhost:3000/a No environment

http://localhost:3000/api/auth/login Save Share

POST http://localhost:3000/api/auth/login Send

Docs Params Auth Headers (8) Body Scripts Settings Cookies

raw JSON Schema Beautify

```
{
  "email": "bhargav@gmail.com",
  "password": "password123"
}
```

Body 200 OK 482 ms 1.11 KB

{ } JSON Preview Visualize


```
{
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
        eyJpZCI6IjQ4ZDE5NzJjLThiM2UtNGQzNy05ZTEwLWNkYTg1NmY5MjMxZSIsInJvbGUiOiJsaWJy
        YXJpYW4iLCJpYXQiOjE3NzY1MjQ3NTksImV4cCI6MTc3OTExNjc1OX0.
        CrZ4ZjUXB8SUhfORD_WD7zW3stknSsFxVE0bKV6g76I"
}
```

POST http://localhost:300 | POST http://localhost:300 | POST http://localhost:300 | No environment

http://localhost:3000/api/books

Save | Share

POST http://localhost:3000/api/books?a Send

Docs | Params | Auth | Headers (9) | Body | Scripts | Settings | Cookies

☑	Host	<calculated when request is sen...
☑	User-Agent	PostmanRuntime/7.51.1
☑	Accept	*/*
☑	Accept-Encoding	gzip, deflate, br
☑	Connection	keep-alive
☑	Authorization	Bearer eyJhbGciOiJIUzI1NiIsInR...

Body | 201 Created | 31 ms | 1.1 KB

JSON | Preview | Visualize


```
{
    "id": "0c08658d-29ae-4406-955f-ddccd624a013",
    "title": "Clean Code",
    "author": "Robert C. Martin",
    "isbn": "9780132350884",
    "publishedYear": 2008,
    "totalCopies": 5,
    "availableCopies": 5,
    "createdAt": "2026-04-18T15:12:08.157Z"
}
```


POST http://localhost:300 | POST http://localhost:300 | POST http://localhost:300 | No environment

http://localhost:3000/api/books

Save | Share

POST http://localhost:3000/api/books?a Send

Docs | Params | Auth | Headers (9) | Body | Scripts | Settings | Cookies

raw | JSON | Schema | Beautify


```
  "title": "Clean Code",
  "author": "Robert C. Martin",
  "isbn": "9780132350884",
  "publishedYear": 2008,
  "totalCopies": 5
}
```


Body | 201 Created | 31 ms | 1.1 KB

JSON | Preview | Visualize


```
{
    "id": "0c08658d-29ae-4406-955f-ddccd624a013",
    "title": "Clean Code",
    "author": "Robert C. Martin",
    "isbn": "9780132350884",
    "publishedYear": 2008,
    "totalCopies": 5,
    "availableCopies": 5,
    "createdAt": "2026-04-18T15:12:08.157Z"
}
```

POST http://localh • POST http://localh • POST http://localh • POST http://localh • No environment

http://localhost:3000/api/auth/register Save Share

POST http://localhost:3000/api/auth/register Send

Docs Params Auth Headers (8) Body Scripts Settings Cookies

raw JSON Schema Beautify

```
{
  "name": "Bhargav",
  "email": "limbanibhargavmaheshbhai@gmail.com",
  "password": "password123",
  "role": "student"
}
```

Body 201 Created • 242 ms • 1.06 KB

{} JSON Preview Visualize

```
{
    "message": "User registered successfully",
    "user": {
        "id": "62137430-a999-4924-bced-08f45dcbc57c",
        "name": "Bhargav",
        "email": "limbanibhargavmaheshbhai@gmail.com",
        "role": "student"
    }
}
```

POST http://localhost:300 • POST http://localhost:3000/ • POST http://localhost:3000/ • POST http://localhost:3000/ • POST http://localhost:3000/ • No environment

http://localhost:3000/api/auth/login Save Share

POST http://localhost:3000/api/auth/login Send

Docs Params Authorization Headers (8) Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
{
  "email": "limbanibhargavmaheshbhai@gmail.com",
  "password": "password123"
}
```

Body Cookies Headers (19) Test Results 200 OK • 272 ms • 1.1 KB

{} JSON Preview Visualize


```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
    eyJpZCI6IjYyMTM3NDMwLWE5OTktNDkyNC1iY2VkLTA4ZjQ1ZGNiYzU3YyIsInJvbGUiOiJzdHVkZW50IiwiaWF0IjoxNzc2NTI3MDExLCJleHAiOjE3NzkxMTkwMTF9.
    7GWpcXag35Xnb2PRAeJeMUkjguZ-H3K7tk5SZSvSNOQ"
}
```

http://localhost:3000/api/transactions/issue

Save | Share

POST http://localhost:3000/api/transactions/issue Send

Docs | Params | Authorization | Headers (9) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1  {
2    "bookId": "0c08658d-29ae-4406-955f-ddccd624a013",
3    "userId": "62137430-a999-4924-bced-08f45dcbc57c"
4  }
5
```

Body | Cookies | Headers (19) | Test Results | 201 Created • 27 ms • 1.15 KB

{} JSON | Preview | Visualize

```
1  {
2      "id": "90fcdb9c-be41-49f2-b583-e57a5243402c",
3      "bookId": "0c08658d-29ae-4406-955f-ddccd624a013",
4      "userId": "62137430-a999-4924-bced-08f45dcbc57c",
5      "issueDate": "2026-04-18T15:47:19.680Z",
6      "dueDate": "2026-05-02T15:47:19.680Z",
7      "returnDate": null,
8      "status": "issued",
9      "fine": 0
10 }
```

POST http://localhc | POST http://localhc | POST http://localhc | POST http://localhc | POST http://localhc | POST http://localhc | GET http://localhos | No environment

http://localhost:3000/api/reports/inventory

Save | Share

GET http://localhost:3000/api/reports/inventory Send

Docs | Params | Authorization | Headers (7) | Body | Scripts | Settings | Cookies

User-Agent	PostmanRuntime/7.51.1	
Accept	*/*	
Accept-Encoding	gzip, deflate, br	
Connection	keep-alive	
Authorization	Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZC...	
Key	Value	Description

Body | Cookies | Headers (19) | Test Results | 200 OK • 34 ms • 975 B

{} JSON | Preview | Visualize

```
1  {
2      "totalTitles": 1,
3      "totalCopies": 5,
4      "totalAvailable": 4,
5      "totalIssued": 1
6  }
```

Controller.js U | { } users.json U × | JS TransactionUseCases.js U | JS userRoutes.js U | JS userController.js U

data > { } users.json > ...


```json
[
  {
    "id": "48d1972c-8b3e-4d37-9e10-cda856f9231e",
    "name": "Admin Librarian",
    "email": "bhargav@gmail.com",
    "password": "$2b$10$u1RPVYL/c.DbAZtI8OqoKOGnUrOI.nmAQtCAwpyZDzwOAaztnUJ9q",
    "role": "librarian",
    "createdAt": "2026-04-18T15:04:47.395Z"
  },
  {
    "id": "62137430-a999-4924-bced-08f45dcbc57c",
    "name": "Bhargav",
    "email": "limbanibhargavmaheshbhai@gmail.com",
    "password": "$2b$10$LBKuPiVE4KfgIbCIjSZBZeW9HyB58UsMqgGZEPH6H.O2lLMdL9VpK",
    "role": "student",
    "createdAt": "2026-04-18T15:14:54.266Z"
  }
]
```


Controller.js U | { } transactions.json U × | JS TransactionUseCases.js U | JS userRoutes.js U | JS userCont

data > { } transactions.json > ...


```json
[
  {
    "id": "90fcdb9c-be41-49f2-b583-e57a5243402c",
    "bookId": "0c08658d-29ae-4406-955f-ddccd624a013",
    "userId": "62137430-a999-4924-bced-08f45dcbc57c",
    "issueDate": "2026-04-18T15:47:19.680Z",
    "dueDate": "2026-05-02T15:47:19.680Z",
    "returnDate": null,
    "status": "issued",
    "fine": 0
  }
]
```

Controller.js U | { } books.json U | JS TransactionUseCases.js U | JS userRoutes.js U | JS userController.js U

data > { } books.json > ...


```
[
  {
    "id": "0c08658d-29ae-4406-955f-ddccd624a013",
    "title": "Clean Code",
    "author": "Robert C. Martin",
    "isbn": "9780132350884",
    "publishedYear": 2008,
    "totalCopies": 5,
    "availableCopies": 4,
    "createdAt": "2026-04-18T15:12:08.157Z"
  }
]
```